CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

Basics of COA

Lecture No.- 04



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Recap of Previous Lecture



Topic Memory Addressing

Topic Memory Access

Topics to be Covered

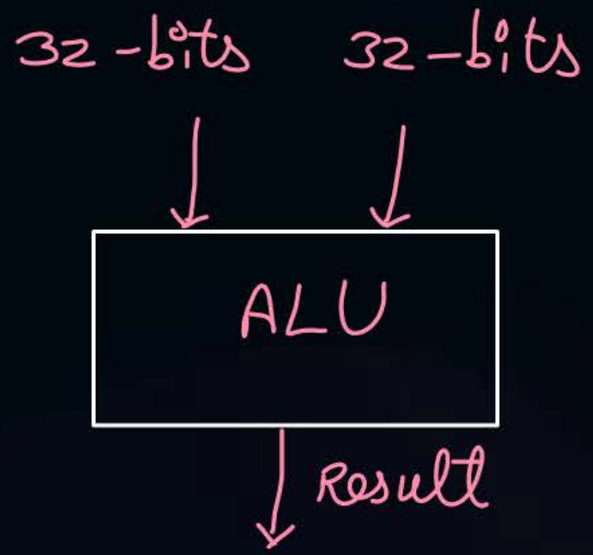


Topic Architecture Type (Based on Size of Input)

Topic Microoperations



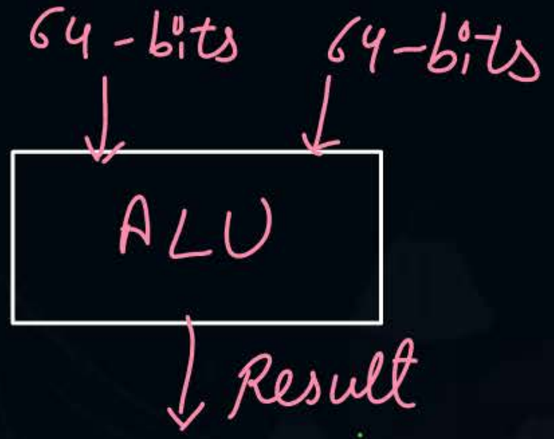
Topic : Architecture Type (Based on Size of Input)



32 -bits architecture

CPU word size = 32 bits
or
4 bytes

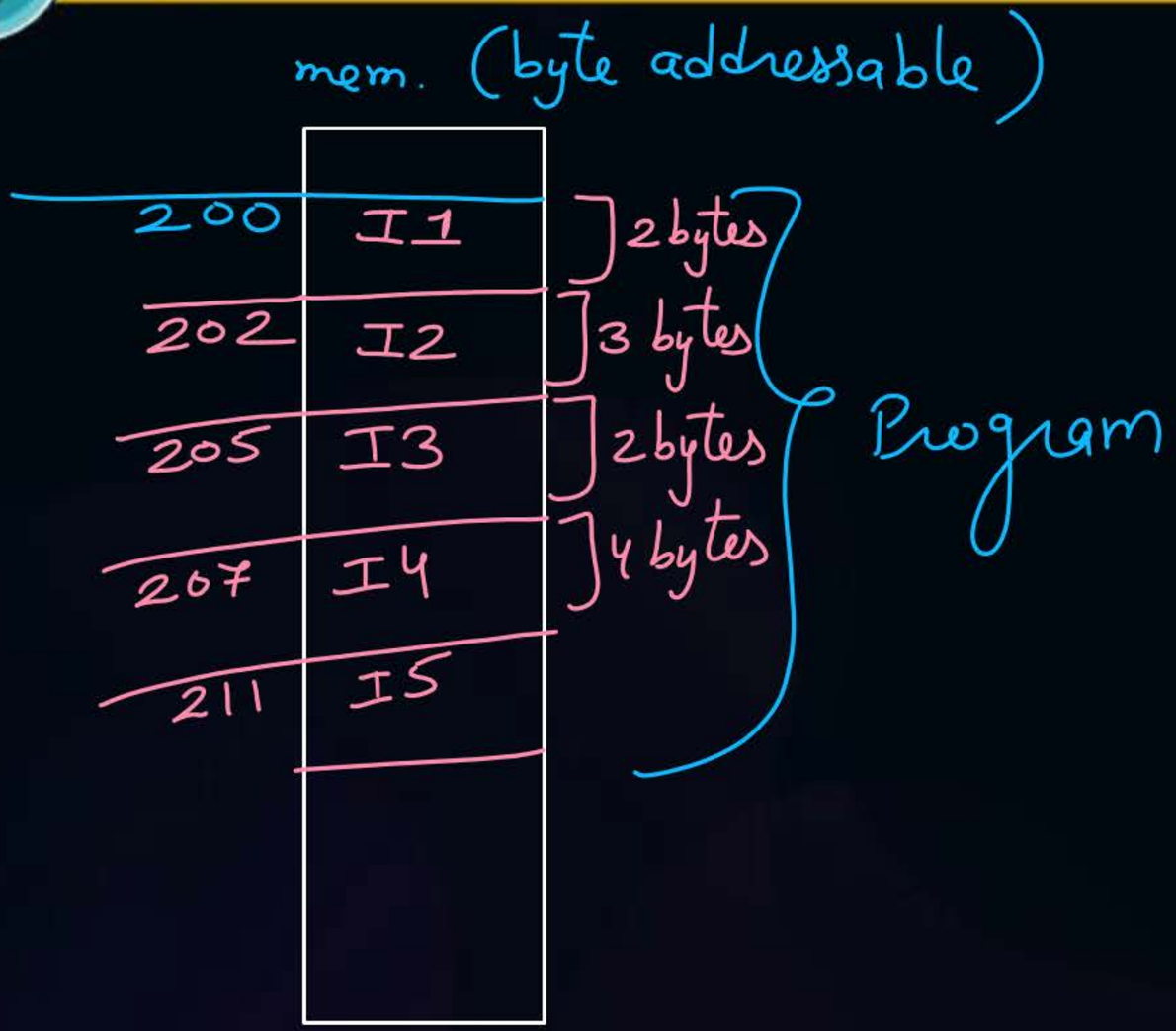
64 - bits architecture



CPU word = 64 bits
or
= 8 bytes

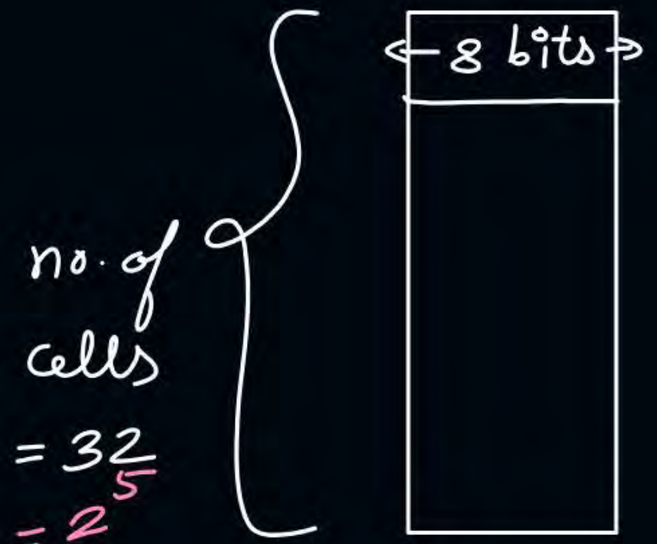


Topic : Address of Variable Size Instructions



memory size and addresses :-

example (byte addressable mem.)



memory size = no. of cells * 1 cell capacity

= 32 * 8 bits

= 256 bits

= 32 bytes

ex:- no. of cells = 256 = 2^8

cell capacity = 16 bits

address size = 8 bits

mem. capacity = 512 bytes

$2^8 * 16$ bits

$2^8 * 2$ bytes

2^9 bytes = 512 bytes

ex:- mem. ⇒ byte addressable

mem. size = 128 bytes

no. of cells = 128 ?

add. size = 7 bits

$$\text{no. of cells} = \frac{\text{mem. capacity}}{\text{1 cell capacity}}$$

$$= \frac{128\,B}{1\,B}$$

$$= 128 = 2^7$$

ex:- byte addressable memory

size = 64 K bytes

address = 16 bits

soln

$$\text{no. of cells} = \frac{64\,KB}{1\,B}$$

$$= 64K = 2^6 * 2^{10}$$

$$= 2^{16}$$

$$\Downarrow$$

add. = 16 bits

ex:- mem size = 256 Mbytes

word addressable

word size = 4 bytes

address size = 26 bits

$$\text{no. of cells} = \frac{256\,MB}{4\,B}$$

$$= \frac{2^8 \cdot 2^{20}}{2^2}$$

$$= 2^{26} \Rightarrow \text{add.} = 26 \text{ bits}$$

Ques) Consider a computer system which uses word addressable memory. The word size is 2 bytes. The computer system has 31 - bits wide address bus. The maximum possible size of memory supported by system 4 GBytes?

Soln

no. of cells $= 2^{31}$

mem. size $= 2^{31} * 2$ bytes

$= 2^{32}$ bytes

$= 2^{30} * 2^2$ B

$= 4$ GB



Topic : Micro Operation

→ smallest operation which CPU can perform at a time.

- The operations executed on values stored in registers
- Symbolic Notation to describe the micro-ops: Register Transfer Language (RTL)



Topic : Micro Operation

- Register Transfer: $R2 \leftarrow R1$ | $R1 \rightarrow R2$
- Comma: $IR \leftarrow DR$, $PC \leftarrow PC + 1$
- Memory Transfer:

→ both will be performed simultaneously

$R1 = 5$

$R2 = 8$

$R2 \leftarrow R1$

⇓

$R1 = 5$

$R2 = 5$

Read :-

To CPU ← from memory

$DR \leftarrow M[address]$

$DR \leftarrow M[1000]$

$DR \leftarrow M[AR]$

write :-

$M[Address] \leftarrow DR$

ex :-

R1 = ~~5~~ 18

R2 = 8

R3 = ~~10~~ 2

R1 ← R2 + 10

R3 ← R3 − R2

ex :

R2 = ~~5~~ 12

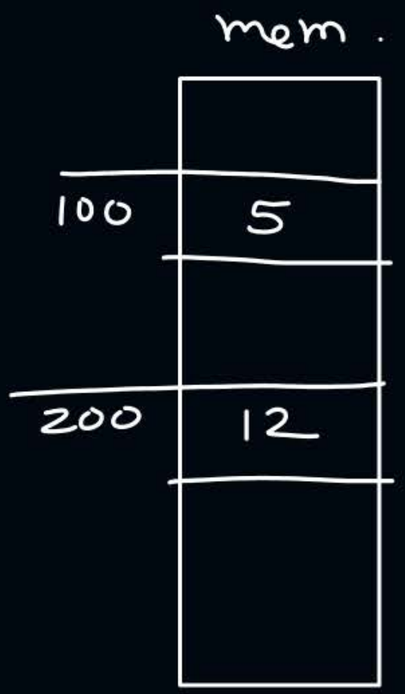
R1 = ~~8~~ ~~13~~ 1

R2 ← M[100]

R1 ← R1 + R2

R2 ← M[200]

R1 ← R1 − R2



esc :-

R1 = 100

R2 = ~~9~~ 21

R3 = 12

$R3 \leftarrow M[R1]$

$R2 \leftarrow R2 + R3$

$M[200] \leftarrow R2$



[NAT]

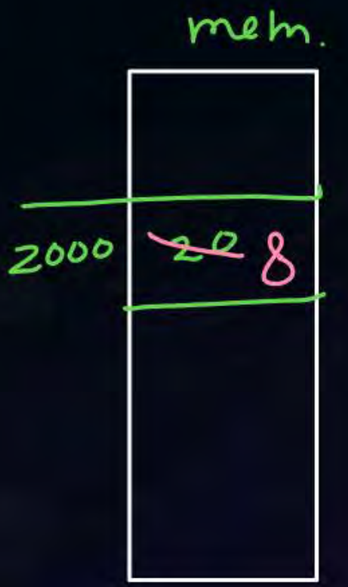
Ans = 8

#Q. Consider the following program segment. Here R1 and R2 are the general purpose register. Assume that the content of memory location 2000 is 20. All numbers are in decimal. After the execution of this program the value of memory location 2000 is?

Instructions	Operations
MOV R1, #12	R1 ← #12
MOV R2, (2000)	R2 ← M[2000]
SUB R2, R1	R2 ← R2 – R1
MOV (2000), R2	M[2000] ← R2
HALT	Stop

R1 = 12

R2 = ~~20~~ 8



[NAT]

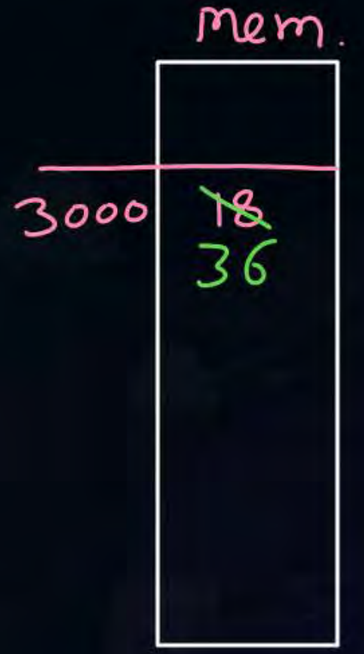
Ans = 36

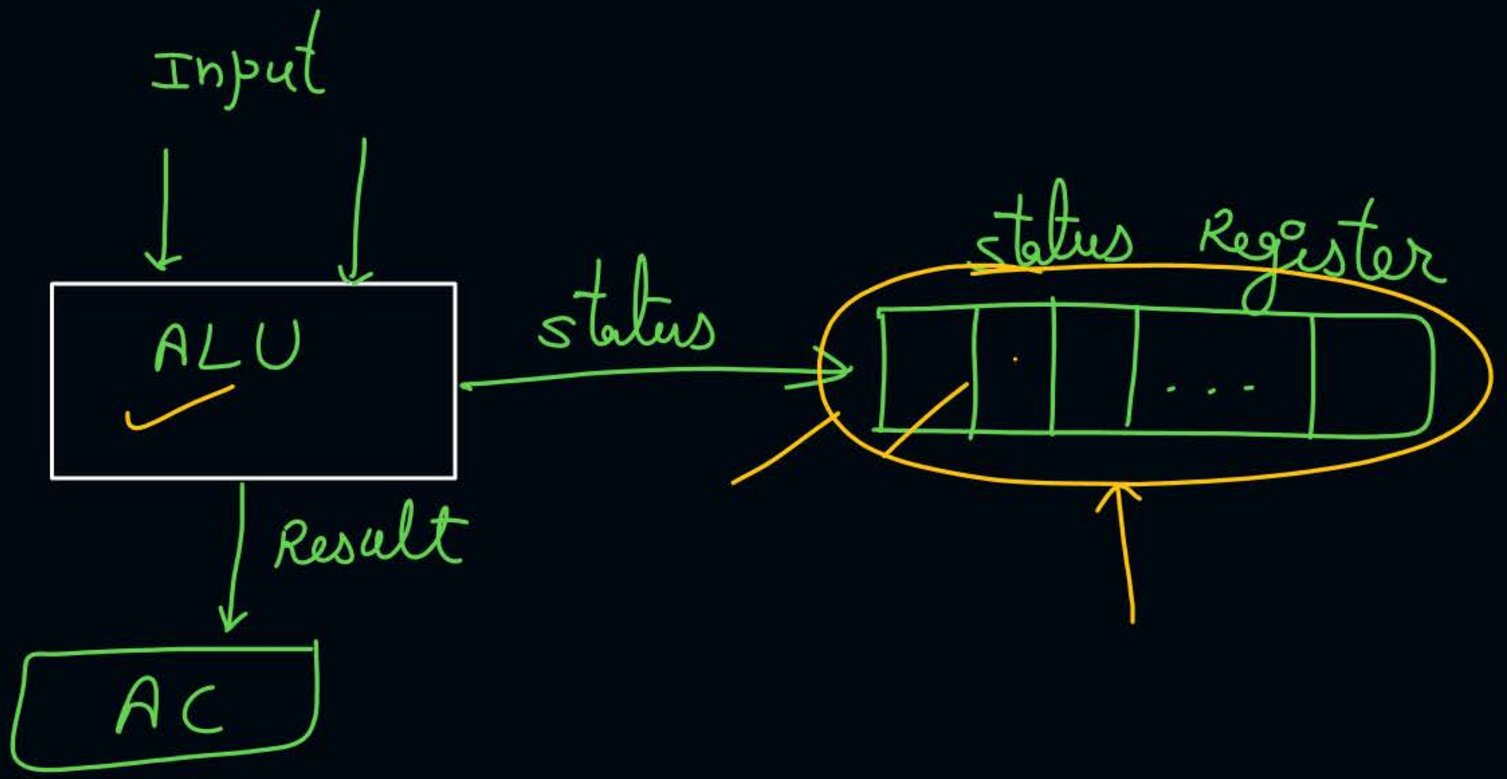
#Q. Consider the following program segment. Here R1 and R2 are the general purpose register. Assume that the content of memory location 3000 is 18. All numbers are in decimal. After the execution of this program the value of memory location 3000 is?

Instructions	Operations
MOV R1, #9	R1 ← #9
MOV R2, (3000)	R2 ← M[3000]
ADD R2, R1	R2 ← R2 + R1
ADD R1, R2	R1 ← R1 + R2
MOV (3000), R1	M[3000] ← R1
HALT	Stop

R1 = ~~9~~ 36

R2 = ~~18~~ 27



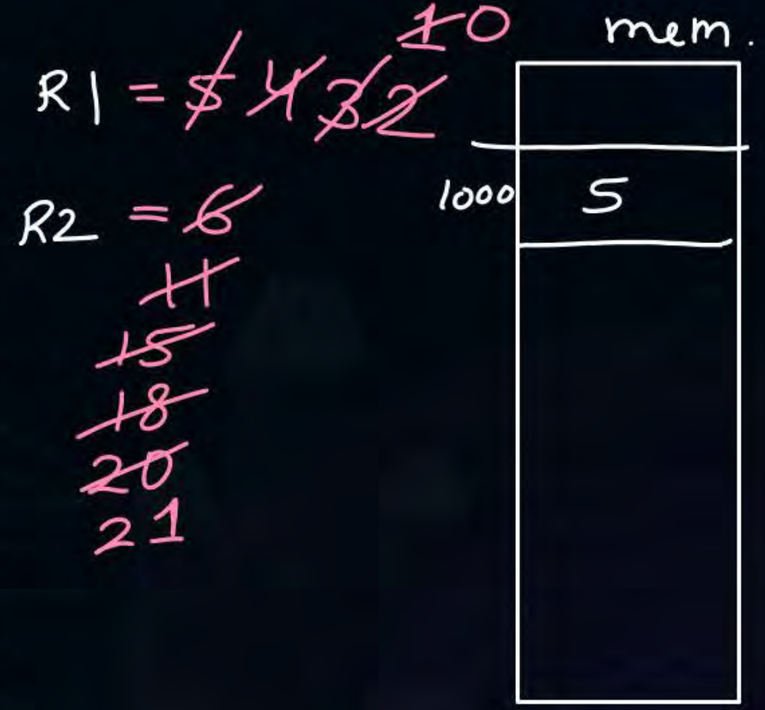

Input
ALU
status
status Register
Result
AC

[NAT]

Ans = 21

#Q. Consider the following program segment. Here R1 and R2 are the general-purpose register. Assume that the content of memory location 1000 is 5. All the numbers are in decimal.

	Instructions	Operations
	MOV R1, (1000)	R1 ← M[1000]
	MOV R2, #6	R2 ← #6
LOOP:	ADD R2, R1	R2 ← R2 + R1
	DEC R1	R1 ← R1 – 1
	BNZ LOOP	Branch on not zero
	HALT	Stop



The value of R2 at the end of program execution is?

[MCQ]

GATE- 2007

#Q. Consider the following program segment. Here R1, R2 and R3 are the general purpose registers.

	Instruction	Operation	Instruction Size (no. of words)
	MOV R1, (3000)	R1 ← M[3000]	2
LOOP:	MOV R2, (R3)	R2 ← M[R3]	1
	ADD R2,R1	R2 ← R1 + R2	1
	MOV (R3),R2	M [R3] ← R2	1
	INC R3	R3 ← R3 + 1	1
	DEC R1	R1 ← R1 – 1	1
	BNZ LOOP	Branch on not zero	2
	HALT	Stop	1

R1 = ~~10~~ ~~9~~ ~~8~~ 7

R2 = ~~100~~ ~~100~~ ~~100~~
~~110~~ ~~109~~ 108

R3 = ~~2000~~ ~~2002~~
~~2001~~ 2003 ...

if loop runs x times

then no. of mem. references = 1 + 2 * x

x = 10

= 1 + 2 * 10

= 21

1000	Instns
2000	~~100~~ 110
2001	~~100~~ 109
2002	~~100~~ 108
2003	~~100~~ 107
2004	~~100~~ 106
2005	~~100~~ 105
2006	~~100~~ 104
2007	~~100~~ 103
2008	~~100~~ 102
2009	~~100~~ 101
2010	100
3000	10

Assume that the content of memory location 3000 is 10 and the content of the register R3 is 2000. The content of each of the memory locations from 2000 to 2010 is 100. The program is loaded from the memory location 1000. All the numbers are in decimal.

Assume that the memory is word addressable. The number of memory reference for accessing the data in executing the program completely is

A 10

B 11

C 20

D ✓ 21

[MCQ]

GATE- 2007

#Q. Consider the following program segment. Here R1, R2 and R3 are the general purpose registers.

	Instruction	Operation	Instruction Size (no. of words)
	MOV R1, (3000	R1 ← M[3000]	2
LOOP:	MOV R2, (R3)	R2 ← M[R3]	1
	ADD R2,R1	R2 ← R1 + R2	1
	MOV (R3),R2	M [R3] ← R2	1
	INC R3	R3 ← R3 + 1	1
	DEC R1	R1 ← R1 – 1	1
	BNZ LOOP	Branch on not zero	2
	HALT	Stop	1

Assume that the content of memory location 3000 is 10 and the content of the register R3 is 2000. The content of each of the memory locations from 2000 to 2010 is 100. The program is loaded from the memory location 1000. All the numbers are in decimal.

Assume that the memory is word addressable. After the execution of this program, the content of memory location 2010 is:

A ✓ 100

B 101

C 102

D 110



2 mins Summary

Topic — Architecture Type (Based on Size of Input)

Topic — Microoperations

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Happy Learning

THANK - YOU